DEVOPS CAPSTONE · SESSION 21

seatlock.

An event-ticketing app where two people can never book the same seat, shipped through a production-style DevOps platform.

Sanaa Ara

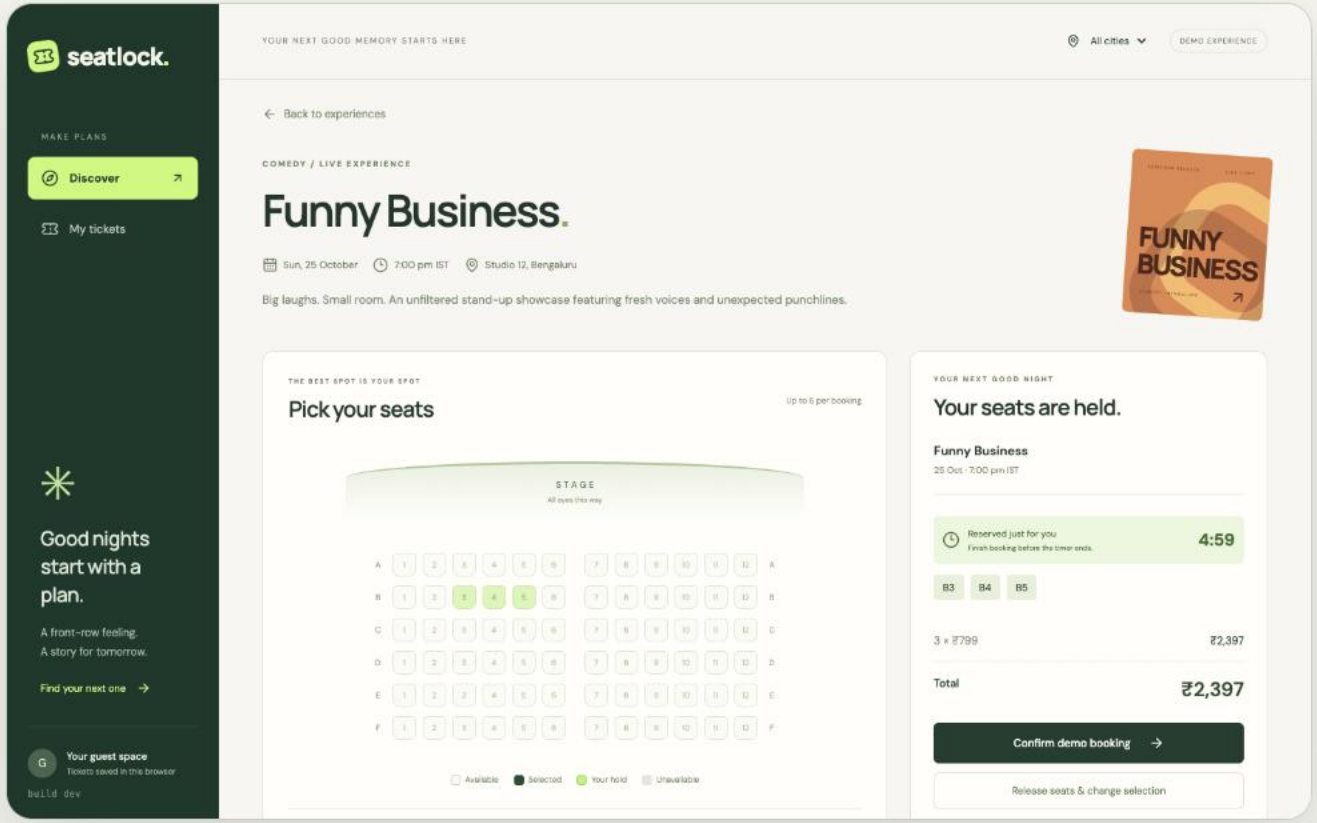
github.com/sanaa-duhh/Seatlock_Devops

01 · OVERVIEW

Two buyers. One seat. Never a double-booking.

- Pick up to six seats; they're held for five minutes
- PostgreSQL row locks decide who owns a seat, on every API replica
- Holds are all-or-nothing; confirming twice never books twice

16 endpoints · 29 tests · 88% coverage



Seats held, with the five-minute countdown running

02 · TECH STACK

One app, eight layers of tooling

APP

FastAPI · React · PostgreSQL · SQLAlchemy · Alembic

TESTING

Pytest · Playwright · Ruff · Prettier

CONTAINERS

Docker multi-stage · Compose · GHCR

CI/CD

GitHub Actions · Dependabot · kind

SECURITY

Trivy · Gitleaks · Bandit · pip-audit · npm audit

INFRASTRUCTURE

Terraform · AWS VPC · EKS · EBS · IAM

KUBERNETES

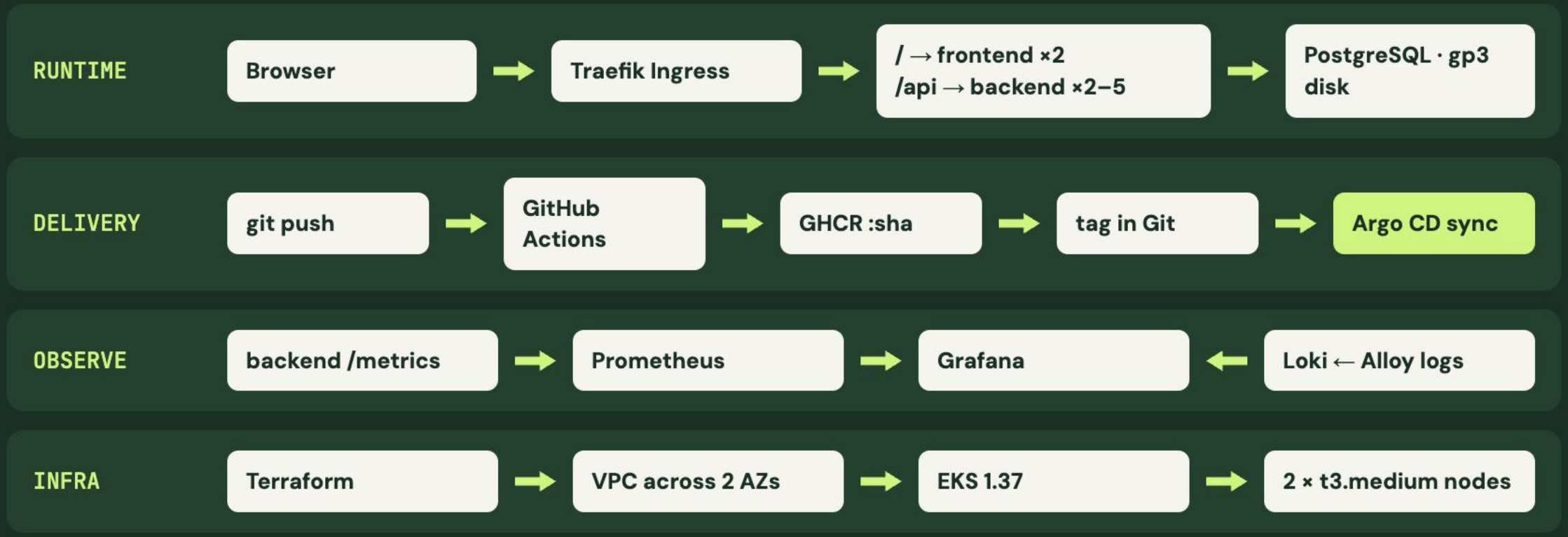
Helm · Argo CD · Traefik Ingress · HPA

OBSERVABILITY

Prometheus · Grafana · Loki · Alloy

03 · ARCHITECTURE

From a git push to a monitored EKS cluster



04 · CI/CD + DEVSECOPS

Ten jobs before anything ships



Security gates. Bandit, pip-audit, npm audit, Gitleaks over the whole Git history, Trivy on both images and the IaC. Any HIGH or CRITICAL fails the run.

Real catches. 46 → 0 HIGH CVEs in the backend image; CVE-2026-4775 in libtiff, hidden by a stale build cache.

.github/workflows/ci.yml · images tagged with the commit SHA · actions pinned to commit SHAs

05 · GITOPS WITH ARGO CD

Git is the deploy button

1

CI pushes the scanned images to GHCR, tagged with the commit SHA

2

CI commits that tag to the GitOps values file

3

Argo CD in the cluster sees Git change

4

It renders the Helm chart and rolls out the new Pods

5

Manual drift is reverted automatically (self-heal)

Pull, not push. The cluster fetches its desired state. CI never holds cluster credentials or AWS keys.

Rollback is a git revert. Every deployed version is a commit with its own history.

gitops/ · Argo CD Application with automated sync, prune and self-heal

06 · INFRASTRUCTURE AS CODE

Terraform builds the whole AWS side

VPC across 2 AZs: 2 public + 2 private subnets

EKS 1.37 with a managed node group, 2 × t3.medium

Add-ons: VPC CNI, CoreDNS, metrics-server, EBS CSI, Pod Identity

Secrets encrypted with a KMS key; nodes run least-privilege IAM roles

BUILT FOR A LOCKED-DOWN ACCOUNT

One region allowed by policy → region is a variable

OIDC providers denied → EKS Pod Identity

Load balancers disabled → ingress_mode = hostport

Found with free dry-run probes before creating anything.

terraform plan: 60 resources · about $0.22 per hour · make eks-down tears it all down

07 · KUBERNETES + HELM

One Helm chart, hardened by default

Resource	What it does
Deployments	frontend ×2, backend ×2–5
HPA	scaled 2 → 4 Pods under load
ClusterIP Services	stable names for each tier
Ingress	/ → frontend, /api → backend
StatefulSet + PVC	PostgreSQL with durable data
Migration Job	Alembic runs per release
ServiceMonitor	tells Prometheus what to scrape

EVERY POD

Runs as non-root

Read-only root filesystem

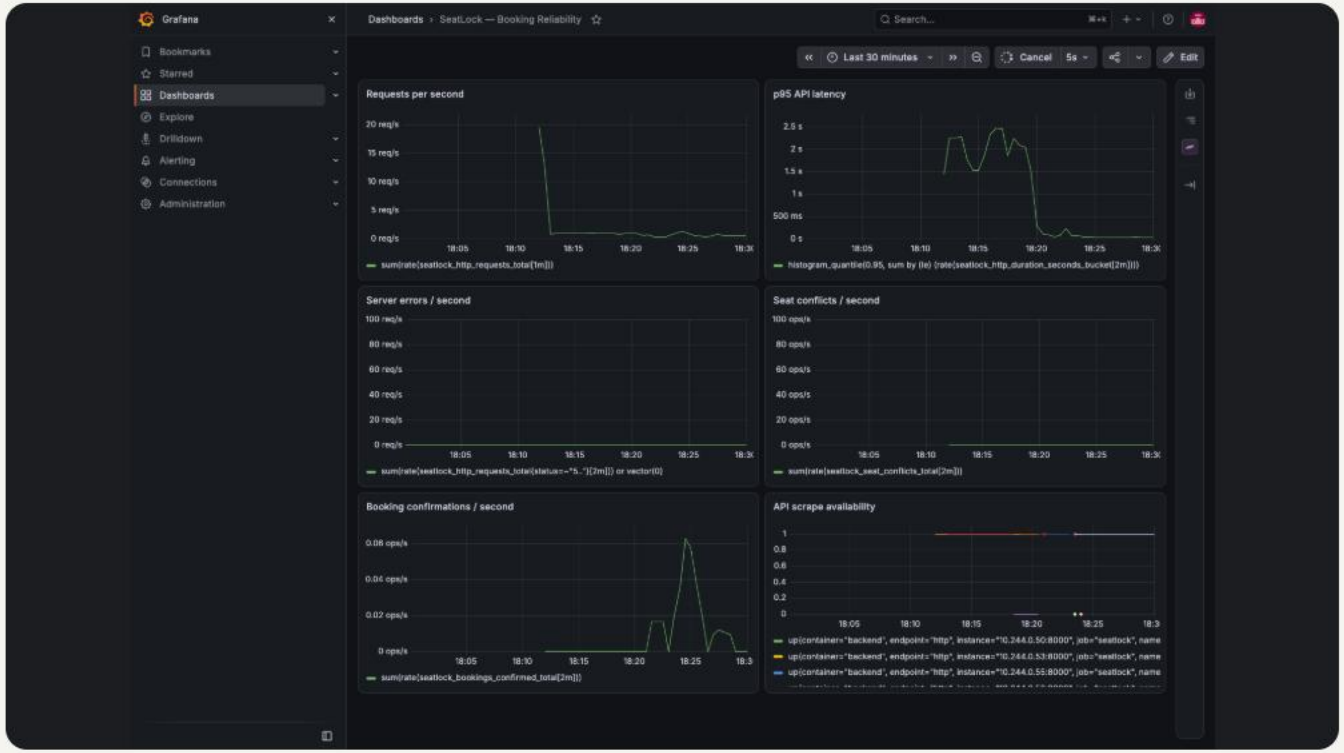
All Linux capabilities dropped

Startup, liveness and readiness probes

CPU and memory requests and limits

helm/seatlock · values.yaml for local, values-eks.yaml for AWS

Metrics and logs you can act on



CUSTOM METRICS

`http_requests_total`
`http_duration_seconds`
`seat_conflicts_total`
`bookings_confirmed_total`

A ServiceMonitor makes Prometheus scrape every backend replica.

Alerts fire when the API is down or returning 5xx.

Alloy ships JSON logs to Loki, each with a request ID.

kube-prometheus-stack · Grafana dashboard: monitoring/dashboards/seatlock.json

09 · KEY LEARNINGS

Five things I'd tell my past self

01	**Correctness lives in the database.** A UI countdown can't stop two replicas selling one seat; row locks can.
02	**A dependency audit is not an image scan.** pip-audit was clean while the image held 46 HIGH CVEs.
03	**A cached image can be a stale image.** Security scans belong on fresh builds.
04	**Test the real network path.** Kubernetes DNS broke what Compose hid, and only a cluster test showed it.
05	**Probe the cloud before you build.** Account guardrails shaped the design, so the code adapts with variables.

Thank you · github.com/sanaa-duhh/Seatlock_Devops